140+ BASIC PYTHON PROGRAMS

This resource can assist you in preparing for your interview

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Program 1

Write a Python program to print "Hello Python".

In [1]:
```
print("Hello Python")
```

```
Hello Python
```

Program 2

Write a Python program to do arithmetical operations addition and division.

In [2]:
```
# Addition
num1 = float(input("Enter the first number for addition: "))
num2 = float(input("Enter the second number for addition: "))
sum_result = num1 + num2
print(f"sum: {num1} + {num2} = {sum_result}")
```

```
Enter the first number for addition: 5
Enter the second number for addition: 6
sum: 5.0 + 6.0 = 11.0
```

In [3]:
```
# Division
num3 = float(input("Enter the dividend for division: "))
num4 = float(input("Enter the divisor for division: "))
if num4 == 0:
    print("Error: Division by zero is not allowed.")
else:
    div_result = num3 / num4
    print(f"Division: {num3} / {num4} = {div_result}")
```

```
Enter the dividend for division: 25
Enter the divisor for division: 5
Division: 25.0 / 5.0 = 5.0
```

Program 3

Write a Python program to find the area of a triangle.

In [4]:
```
# Input the base and height from the user
base = float(input("Enter the length of the base of the triangle: "))
height = float(input("Enter the height of the triangle: "))
# Calculate the area of the triangle
area = 0.5 * base * height
# Display the result
print(f"The area of the triangle is: {area}")
```

```
Enter the length of the base of the triangle: 10
Enter the height of the triangle: 15
The area of the triangle is: 75.0
```

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Program 4

Write a Python program to swap two variables.

In [5]:
```
# Input two variables
a = input("Enter the value of the first variable (a): ")
b = input("Enter the value of the second variable (b): ")
# Display the original values
print(f"Original values: a = {a}, b = {b}")
# Swap the values using a temporary variable
temp = a
a = b
b = temp
# Display the swapped values
print(f"Swapped values: a = {a}, b = {b}")
```

```
Enter the value of the first variable (a): 5
Enter the value of the second variable (b): 9
Original values: a = 5, b = 9
Swapped values: a = 9, b = 5
```

Program 5

Write a Python program to generate a random number.

In [6]:
```
import random
print(f"Random number: {random.randint(1, 100)}")
```

```
Random number: 89
```

Program 6

Write a Python program to convert kilometers to miles.

In [7]:
```
kilometers = float(input("Enter distance in kilometers: "))

# Conversion factor: 1 kilometer = 0.621371 miles
conversion_factor = 0.621371

miles = kilometers * conversion_factor

print(f"{kilometers} kilometers is equal to {miles} miles")
```

```
Enter distance in kilometers: 100
100.0 kilometers is equal to 62.137100000000004 miles
```

Program 7

Write a Python program to convert Celsius to Fahrenheit.

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In [8]:
```
celsius = float(input("Enter temperature in Celsius: "))

# Conversion formula: Fahrenheit = (Celsius * 9/5) + 32
fahrenheit = (celsius * 9/5) + 32

print(f"{celsius} degrees Celsius is equal to {fahrenheit} degrees Fahr
```

```
Enter temperature in Celsius: 37
37.0 degrees Celsius is equal to 98.6 degrees Fahrenheit
```

Program 8

Write a Python program to display calendar.

In [9]:
```
import calendar

year = int(input("Enter year: "))
month = int(input("Enter month: "))

cal = calendar.month(year, month)
print(cal)
```

```
Enter year: 2023
Enter month: 11
   November 2023
Mo Tu We Th Fr Sa Su
       1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30
```

Program 9

Write a Python program to solve quadratic equation.

The standard form of a quadratic equation is:

$$ax^2 + bx + c = 0$$

where

a, b and c are real numbers and

$a \neq 0$

The solutions of this quadratic equation is given by:

$$(-b \pm (b^2 - 4ac)^{1/2})/(2a)$$

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In [10]:

```
import math

# Input coefficients
a = float(input("Enter coefficient a: "))
b = float(input("Enter coefficient b: "))
c = float(input("Enter coefficient c: "))

# Calculate the discriminant
discriminant = b**2 - 4*a*c

# Check if the discriminant is positive, negative, or zero
if discriminant > 0:
    # Two real and distinct roots
    root1 = (-b + math.sqrt(discriminant)) / (2*a)
    root2 = (-b - math.sqrt(discriminant)) / (2*a)
    print(f"Root 1: {root1}")
    print(f"Root 2: {root2}")
elif discriminant == 0:
    # One real root (repeated)
    root = -b / (2*a)
    print(f"Root: {root}")
else:
    # Complex roots
    real_part = -b / (2*a)
    imaginary_part = math.sqrt(abs(discriminant)) / (2*a)
    print(f"Root 1: {real_part} + {imaginary_part}i")
    print(f"Root 2: {real_part} - {imaginary_part}i")

```

```
Enter coefficient a: 1
Enter coefficient b: 4
Enter coefficient c: 8
Root 1: -2.0 + 2.0i
Root 2: -2.0 - 2.0i
```

Program 10

Write a Python program to swap two variables without temp variable.

In [11]:

```
a = 5
b = 10

# Swapping without a temporary variable
a, b = b, a


print("After swapping:")
print("a =", a)
print("b =", b)

```

```
After swapping:
a = 10
b = 5
```

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